Md Rayhanul Masud

PhD in Computer Science, University of California, Riverside

Open to remote or on-site in Southern California

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Summary

PhD in Computer Science at UC Riverside with two years of software engineering experience, working at the intersection of machine learning, cybersecurity, and large-scale software ecosystems (GitHub, online forums). Built ML-driven systems for repository search, cross-platform identity linkage, and malware ecosystem analysis using technical and social signals. Open to research and applied ML roles where rigorous, production-minded work drives real-world impact.

Education

University of California, Riverside — Sep 2021 – Jun 2026
Ph.D. in Computer Science — Advisor: Prof. Michalis Faloutsos

University of California, Riverside — Sep 2021 – Dec 2024
M.S. in Computer Science — CGPA: 3.90/4.00

Bangladesh University of Engineering and Technology (BUET) — Apr 2012 – Feb 2017
B.Sc. in Computer Science and Engineering — Advisor: Prof. Md Monirul Islam

Research & Engineering Experience

Graduate Student Researcher — Sep 2021 – Jun 2026
University of California, Riverside

- Designed and implemented RepoScope, an ML-powered “search by example” system for GitHub repositories, combining interpretable metadata features with code embeddings and visual clustering to enable multi-granularity exploration of large code corpora. **RepoScope** improved Top-5 precision by 20% vs baselines in our evaluation.
- Developed **GeekMAN** to match stylized usernames across security forums, GitHub, and social networks, using string-similarity features achieving up to 86% precision on technogeek dataset.
- Developed **MAGNET** to characterize software engineering perspective for malicious malware repository development in GitHub.
- Built reproducible ETL pipelines for GitHub + forum data, defining feature schemas/mapping logic and using SQL/PySpark + Python to validate joins and produce analysis-ready tables and model inputs.

Software Engineer — Jul 2018 – Jul 2020
Kona Software Lab Ltd., Bangladesh

- Developed the transaction management pipeline for the **Kona Blockchain Platform** built on Ethereum; implemented a template-based smart-contract deployment mechanism that reduced insecure transaction requests from client applications by incorporating authorized smart contract interaction.
- Collaborated in an agile, product-focused team to prototype scalable blockchain solutions, contributing to design, implementation, testing, and deployment.

Technical Skills

Languages Python, Java, JavaScript, C++

Data / SQL SQL (MySQL), NoSQL

Data / ML PyTorch, Pandas, PySpark, scikit-learn

Deployment / Cloud Docker, Git, Google Cloud

Selected Publications

Journal & Conference Papers

- **M. R. Masud**, B. Treves, M. Faloutsos. "Disambiguating usernames across platforms: the GeekMAN approach." *Social Network Analysis and Mining (SNAM)*, 2024. [Link]
- **M. R. Masud**, M. O. F. Rokon, Q. Zhang, M. Faloutsos. "MetaSim: A search engine for finding similar GitHub repositories." *International Conference on Software Maintenance and Evolution (ICSME)*, 2024. [Link] [Demo] [Platform]
- **M. R. Masud**, Q. Zhang, M. Faloutsos. "MAGNET: A software-engineering view of the MAlware ecosystem on Github NETwork." *ASONAM*, 2026.
- N. A. Tania, **M. R. Masud**, M. O. F. Rokon, Q. Zhang, M. Faloutsos. "Who is Creating Malware Repositories on GitHub and Why?" *The Web Conference (WWW)*, 2024. [Link]
- **M. R. Masud**, B. Treves, M. Faloutsos. "GeekMAN: Geek-oriented username Matching Across online Networks." *ASONAM*, 2023. [Link] [Presentation]
- **M. R. Masud**, Q. Zhang, M. Faloutsos. "RepoScope: A Multi-Granularity Query-by-Example Framework for GitHub Repository Search." *Under Review.*
- **M. R. Masud**, A. T. Wasi, M. R. Parvez . "Reward Engineering for Reinforcement Learning in Software Tasks." *Under Review.* [Link]

Selected Academic Course Projects

Adversarial Prompt Generation for LLMs [PDF]

- Employed a Morse-code-embedded adversarial prompt generation approach to evaluate safety alignment robustness of LLaMA-2-7B-chat, achieved an 82.1% success rate on the AdvBench dataset.
- Implemented the full experimental pipeline for data generation, prompt engineering, and evaluation.

CalcGPT: Evaluating LLMs on Arithmetic [Code]

- Implemented an evaluation framework to compare GPT-Neo-2.7B and LLaMA-7B on arithmetic tasks.
- LLaMA-7B is found to be more accurate than GPT-Neo-2.7B for adding numbers with 86.1% accuracy.

Search Engine for Twitter Data [PDF]

- Designed and implemented a Twitter search engine on 2.5GB of data, benchmarking Hadoop vs. Lucene retrieval. Lucene built inverted indexes in $\sim$ 10.7–12.0s vs. Hadoop 38.6–140.0s (10k–50k tweets), while evaluating indexing efficiency, query latency, and relevance across configurations.

Popularity & Influence on GitHub [Code]

- Built a user-follower network of 3.5K GitHub users and applied the HITS algorithm to model influence.
- Analyzed how author and repository metadata correlate with network-wide popularity and influence.

Service & Awards

Service

- External Reviewer: ACL ARR 2025; Judge: Cutie Hack 2025; Student Volunteer: KDD 2023.
- Community Contributor: Ethereum Stack Exchange (500+ reputation, $\sim$32K users reached).

Awards

- Dissertation Completion Fellowship Award, University of California, Riverside, 2026.
- Dean's Fellowship Award, University of California, Riverside, 2021.
- Dean's Award (Level-4, Term-1), BUET, 2016.